

Assignment B — HTLC (Hashed Time-Lock Contract)

An **HTLC** (Hashed Time-Lock Contract) lets:

- **Alice** claim BTC from **Bob** *only* if she knows a secret (preimage **S**) before a time limit.
- **Bob** can refund the BTC after the time limit expires.

1. Complete the HTLC script

This is the **locking (redeem) script** that holds the funds.

- OP_IF
- # Alice's branch – can spend with the secret before timeout
- OP_HASH160 <Hash(S)> OP_EQUALVERIFY
- <AlicePubKey> OP_CHECKSIG
- OP_ELSE
- # Bob's refund branch – after 21 minutes
- <LockTime> OP_CHECKLOCKTIMEVERIFY OP_DROP
- <BobPubKey> OP_CHECKSIG
- OP_ENDIF

Explanation

- OP_IF branch: executes if Alice provides **1** on the stack → requires the **secret preimage S** and her **signature**.
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2. Claiming transaction script (Alice)

Alice wants to spend **before 21 minutes**, using her secret S.

Unlocking script (scriptSig / witness):

<signature_Alice> <S> 1

Stack when executed:

1. Push Alice's signature
2. Push secret preimage S
3. Push 1 → chooses the OP_IF branch

3. Refund transaction script (Bob)

Bob wants refund **after 21 minutes**.

Unlocking script:

<signature_Bob> 0

Stack:

1. Push Bob's signature
2. Push 0 → chooses the OP_ELSE branch

4. Test with sample hash and timeout

Let's pick sample values to make it concrete.

- Secret: S = "ILoveBitcoin"
- Hash160(S): 4a5d6e...
- LockTime: 1260 seconds (≈ 21 minutes)

Example HTLC with real numbers

OP_IF

OP_HASH160 4a5d6e2c63a1234b9e1f4b123a8c78d12e3f45ab

OP_EQUALVERIFY

02a1633caf7c5c57c1f... OP_CHECKSIG

OP_ELSE

1260 OP_CHECKLOCKTIMEVERIFY OP_DROP

03b7a24dfd9c91d2a8f... OP_CHECKSIG
OP_ENDIF